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RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Telangana State Weather Data Analysis

Team Members

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Batch No:487

ABSTRACT

- Analyzed weather data of Telangana state from 2021 to 2024
- Built interactive dashboards using Power BI
- Data covers 33 districts and 600+ mandals
- Explored rainfall, temperature, humidity and wind speed patterns

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DATA DESCRIPTION

- District
- Mandal
- Date
- Rain (mm)
- Min Temp (°C)
- Max Temp (°C)
- Min Humidity (%)
- Max Humidity (%)
- Min Wind Speed (Kmph)
- Max Wind Speed (Kmph)
- Season (created by us in Power Query)

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DATA LOADING PROCESS

Data Loading

C:\Users\shara\Downloads\Weather Data

Content	Name	Extension	Date accessed	Date modified	Date created	Attributes	
Binary	TS Weather Data April 2021 .csv	.csv	10-04-2026 16:50:03	11-01-2024 12:40:39	08-04-2026 10:37:18	Record	C:\Users\shai
Binary	TS Weather data August 2021.csv	.csv	10-04-2026 17:03:29	11-01-2024 12:41:23	08-04-2026 10:37:18	Record	C:\Users\shai
Binary	TS Weather data December 2021.csv	.csv	10-04-2026 17:03:29	11-01-2024 12:41:52	08-04-2026 10:37:18	Record	C:\Users\shai
Binary	TS Weather data July 2021.csv	.csv	10-04-2026 17:03:30	11-01-2024 12:41:16	08-04-2026 10:37:18	Record	C:\Users\shai
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Binary	TS Weather data May 2021.csv	.csv	10-04-2026 17:03:30	11-01-2024 12:40:48	08-04-2026 10:37:18	Record	C:\Users\shai
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Binary	TS Weather data December 2022.csv	.csv	09-04-2026 14:05:10	11-01-2024 12:45:34	08-04-2026 10:37:18	Record	C:\Users\shai
Binary	TS Weather data February 2022.csv	.csv	09-04-2026 14:05:11	11-01-2024 12:43:11	08-04-2026 10:37:18	Record	C:\Users\shai
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The data in the preview has been truncated due to size limits.

Combine | Load | Transform Data | Cancel

Data Preparation Steps

- Removed unnecessary columns (Source.Name)
- Appended XLSX file (Jan-Mar 2021) to main table
- Created Season column — Summer, Monsoon, Post-Monsoon, Winter
- Created Daily Temp Range measure — Max Temp minus Min Temp
- Created Daily Wind Range measure — Max Wind minus Min Wind

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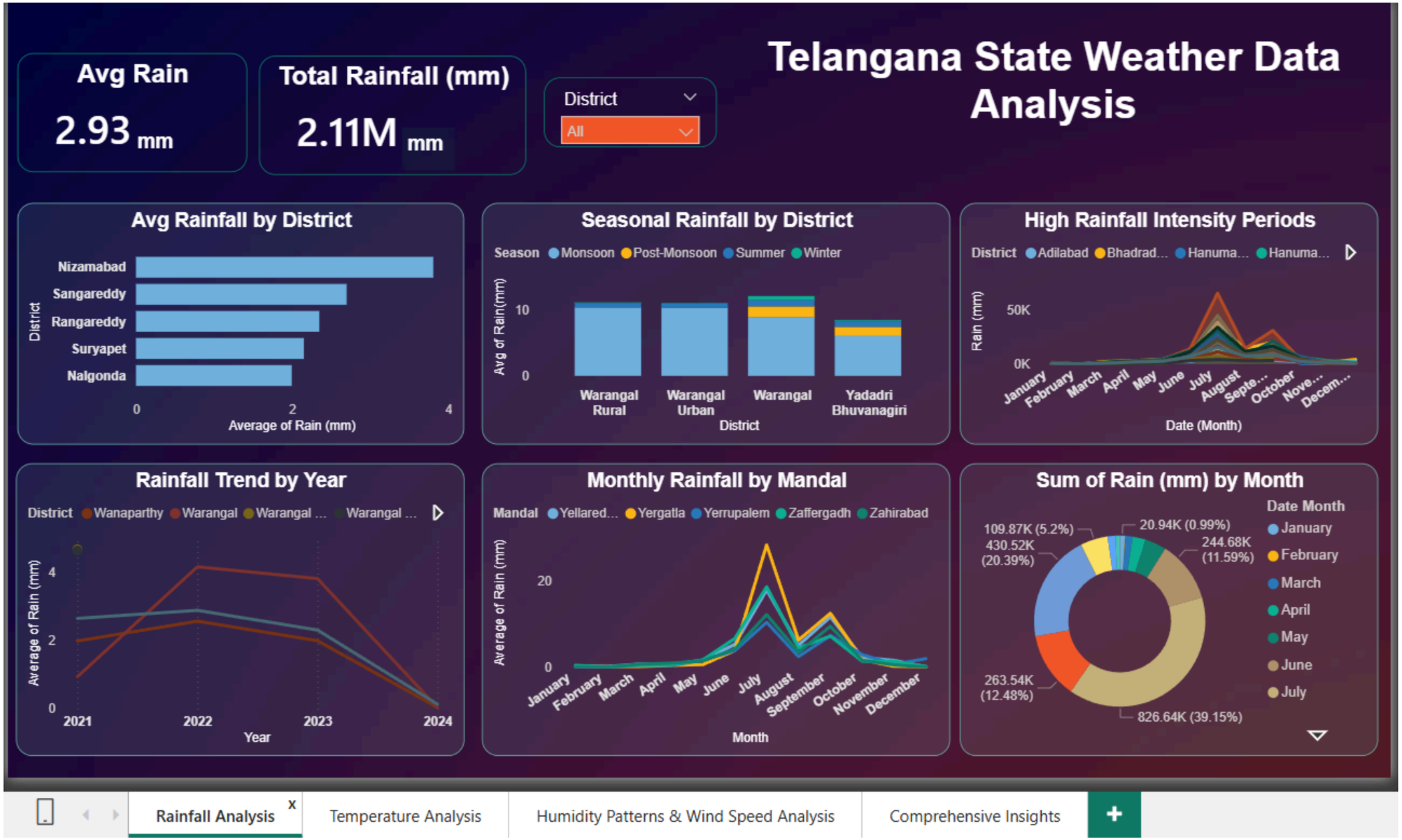
RAINFALL ANALYSIS



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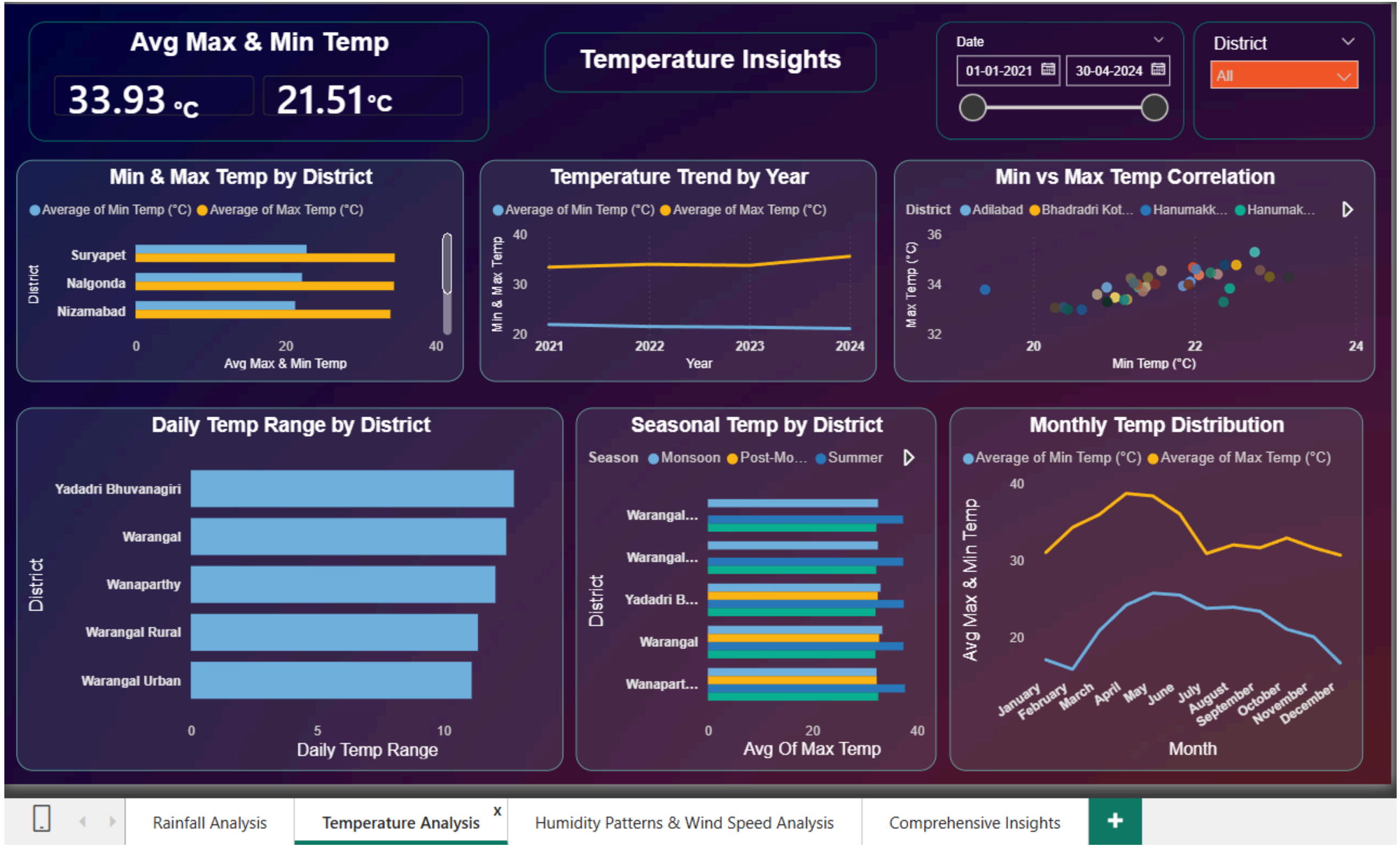
TEMPERATURE ANALYSIS



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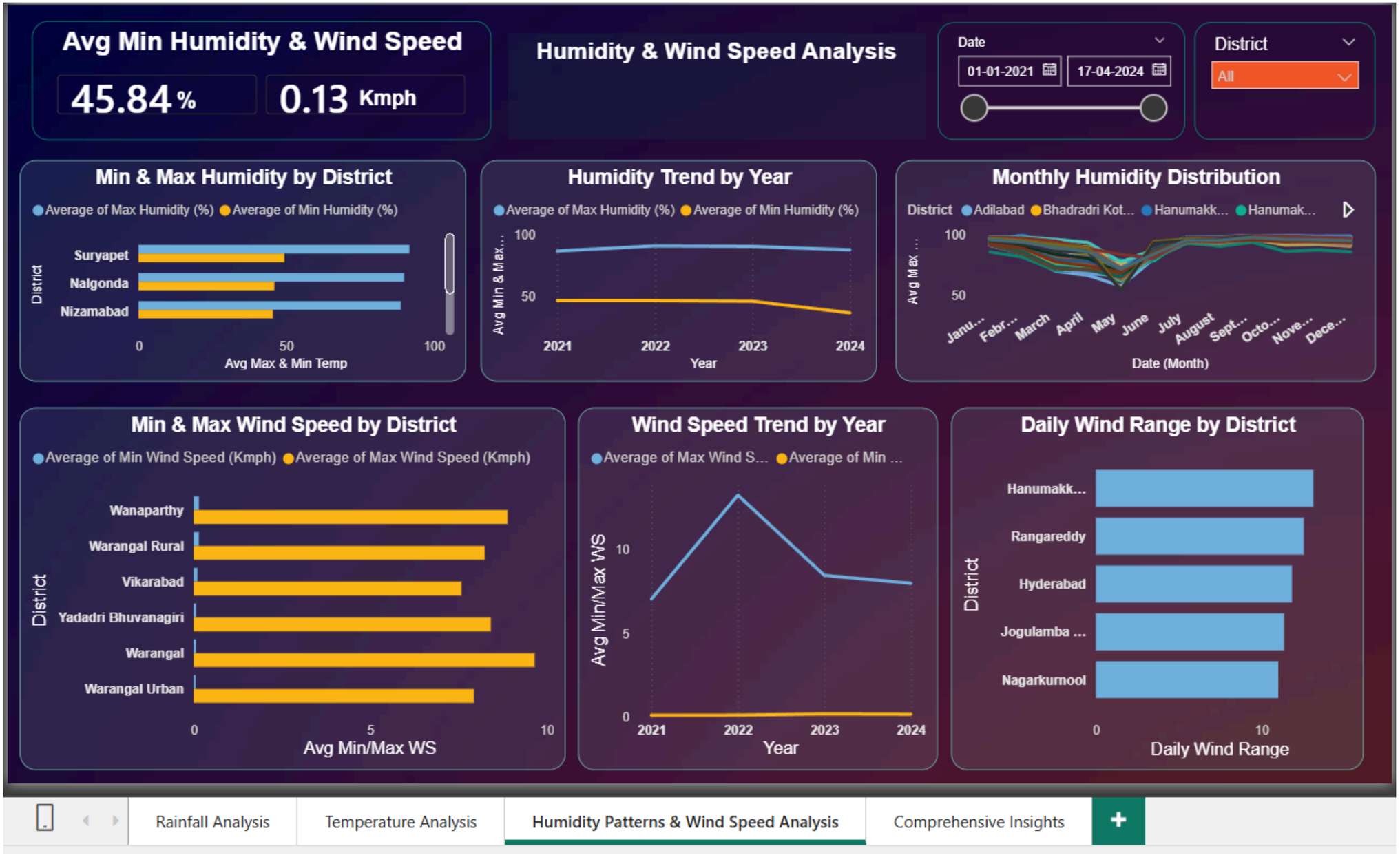
HUMIDITY & WIND SPEED ANALYSIS



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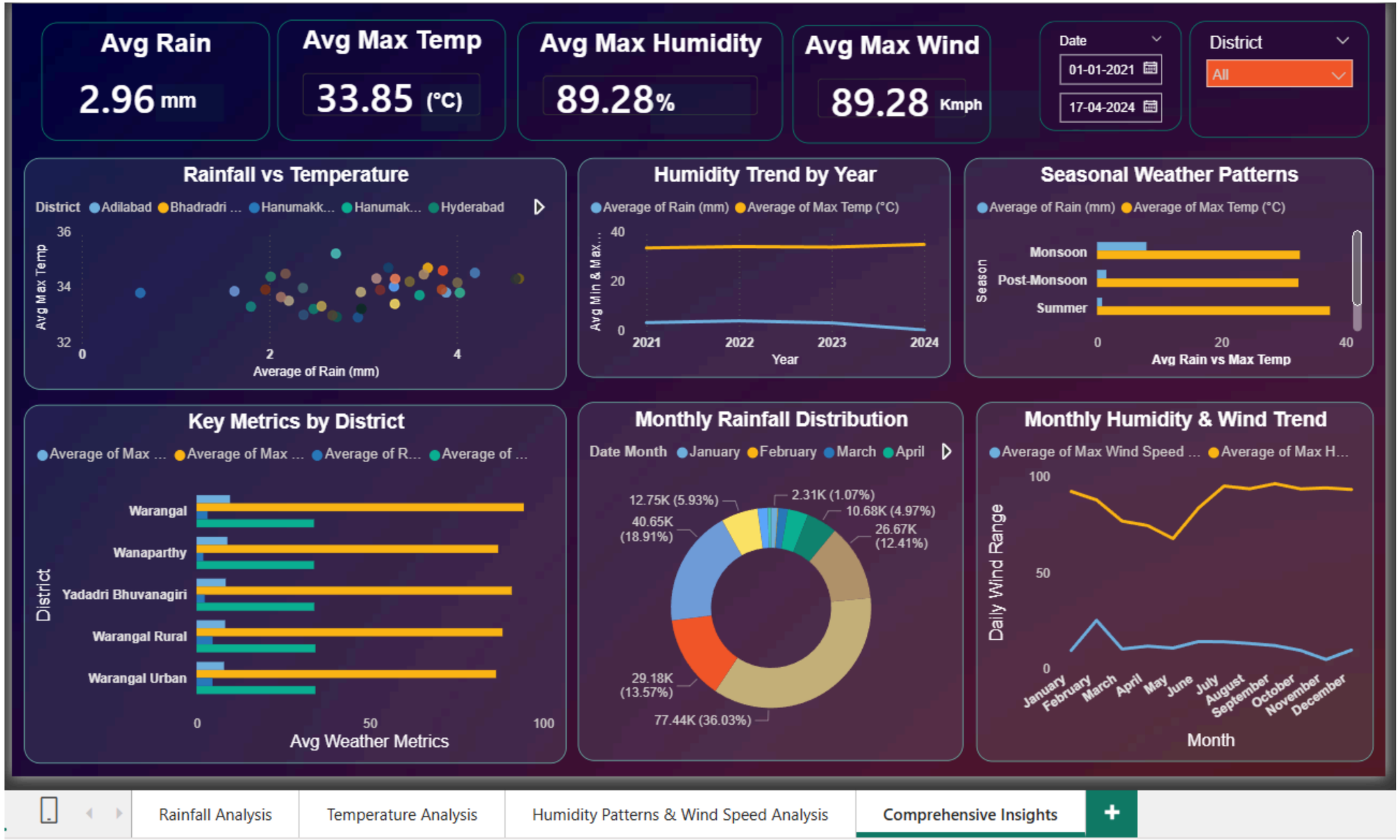
COMPREHENSIVE INSIGHTS



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KEY INSIGHTS

- Warangal Rural & Urban receive highest average rainfall in Telangana
- July alone contributes 39% of annual rainfall peak monsoon month
- Suryapet & Nalgonda have highest maximum temperatures hottest districts
- 2022 was the highest rainfall year across most districts 2024 shows sharp drop
- Wanaparthy has highest wind speeds windiest district in Telangana
- Humidity drops sharply in July despite being peak monsoon interesting pattern
- Monsoon season dominates all weather metrics across all districts

DATA DRIVEN DECISIONS

- Farmers can plan crop seasons based on rainfall patterns
- Government can identify flood-risk districts during July-August
- Disaster management teams can prepare for extreme weather events
- Tourism industry can plan based on seasonal weather patterns

THANK YOU

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